AMMA: AI-ENABLED MATERNAL MONITORING ASSISTANT

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***Abstract*—This paper presents the design and implementation of AMMA, a wearable maternal movement monitoring prototype intended for continuous, home based observation during the third trimester of pregnancy. The current prototype is built around an ESP32-S3 microcontroller interfaced with two MPU6050 inertial sensors and two piezoelectric sensors for detecting abdominal movement, sampled through a sixteen bit ADS1115 analog to digital converter. The system acquires sensor data at ten hertz, displays live readings on an onboard OLED screen, and transmits summarized readings to a ThingSpeak cloud channel every twenty seconds over WiFi. Bench testing confirmed a functioning end to end pipeline from the sensors through the microcontroller to cloud storage, with consistent successful HTTP responses and steadily incrementing entry identifiers on the cloud channel. This work reports the hardware architecture, firmware acquisition loop, and verified cloud integration achieved so far, and identifies signal processing and artificial intelligence based kick classification as the next stage of development.**

***Index Terms*—wearable sensors, maternal health monitoring, fetal movement, Internet of Things, ESP32, accelerometer, piezoelectric sensor, cloud data logging**

I. Introduction

Fetal movement counting is a widely recommended, low cost method for expectant mothers to informally track fetal well being, but conventional kick counting relies on subjective manual perception and is prone to inconsistency and delayed detection of abnormal patterns. AMMA, short for the Maternal Wellness Suite, is being developed as a non invasive wearable belt that replaces subjective counting with objective, sensor based monitoring of maternal and abdominal movement, with continuous home based use as the eventual goal. The broader project scope also includes a mobile application for visualization and an artificial intelligence layer for detecting abnormal movement patterns, but the present paper is scoped to the current working prototype: the sensing hardware, the embedded acquisition firmware, and the cloud logging pipeline that have been implemented and verified to date.

II. Related Work

Wearable and sensor based approaches to fetal and maternal movement monitoring have been explored using a range of sensing modalities. Accelerometer based recognition of fetal activity has been investigated as an objective alternative to manual counting [1], and dedicated signal processing algorithms have been proposed for remote fetal activity monitoring systems [2]. More recent work has combined wearable sensing hardware with cloud or mobile connectivity for maternal and fetal health monitoring [3], including IoT based wearable belts specifically designed to track fetal kicks [4]. Dual accelerometer configurations paired with machine learning classifiers have also been reported for fetal movement detection [5], and earlier prototypes demonstrated remote fetal monitoring during pregnancy using wearable textile and embedded systems [6]. Flexible, skin attachable sensing platforms [7] and vibrational sensor simulation systems [8] further illustrate the range of hardware approaches explored in this space. AMMA builds on these directions by combining dual inertial sensors with dual piezoelectric sensors on a single low cost microcontroller platform with immediate cloud logging, as described in the following sections.

III. System Architecture

The prototype is built around an ESP32-S3 microcontroller, which serves as the central processing and connectivity unit for all sensors, the local display, and the WiFi link to the cloud. Fig. 1 shows the assembled breadboard prototype, with the two inertial sensor modules mounted on opposite sides of the board and the OLED display used for local status monitoring.

Fig. 2 summarizes the overall system workflow: the four sensing elements feed the ESP32-S3 over I^2C, the microcontroller runs a fixed rate acquisition loop, and the resulting readings are simultaneously pushed to the local display, the serial console, and the ThingSpeak cloud channel.

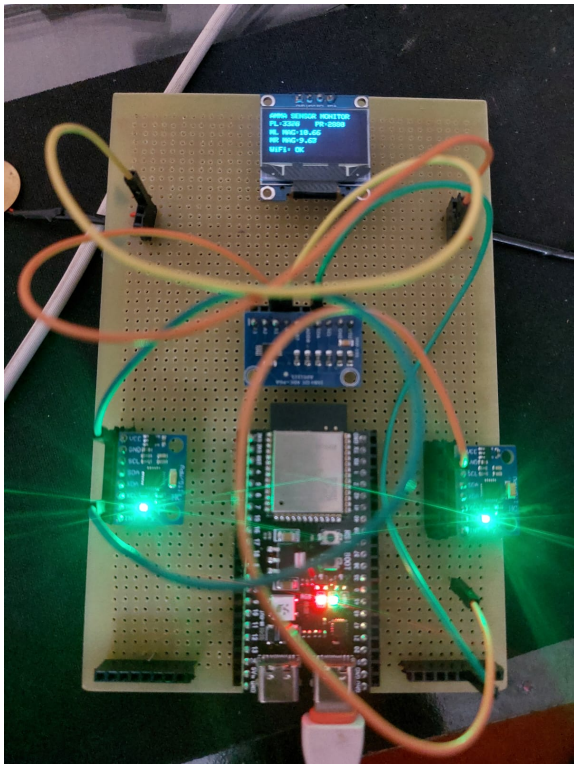


Fig. 1. Assembled AMMA sensor prototype showing the ESP32-S3, the ADS1115 ADC, the two MPU6050 modules (left and right), and the onboard OLED status display.

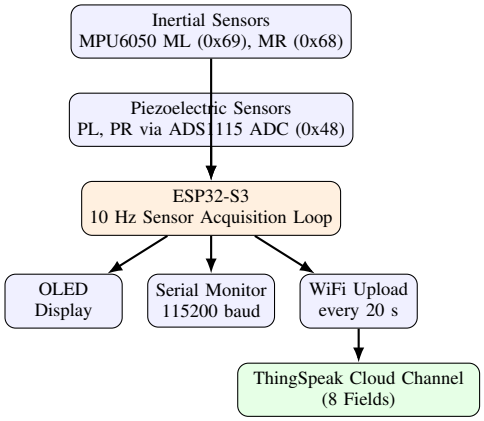


Fig. 2. AMMA system data workflow, from the four sensing elements through the ESP32-S3 acquisition loop to local feedback and cloud logging.

A. Sensing Elements

Table I lists the sensing and display components used in the current prototype.

Two spatially separated sensing regions are used rather than a single sensor pair: the left inertial sensor is designated ML and the right MR, while the left and right piezoelectric elements are designated PL and PR. All devices communicate with the ESP32-S3 over a shared I^{2}C bus running at 400 kHz, with SDA on GPIO 8 and SCL on GPIO 9. Table II lists the I^{2}C address assigned to each device, which allows all four sensing modules and the display to coexist on the same bus.

TABLE I
HARDWARE COMPONENTS OF THE CURRENT PROTOTYPE

Component	Qty	Purpose
MPU6050	2	Abdominal acceleration sensing
Piezoelectric sensor	2	Mechanical/vibration sensing
ADS1115	1	16-bit ADC for piezo signals
OLED display	1	Local status monitoring
ESP32-S3	1	Main controller and WiFi link

TABLE II
I^{2}C DEVICE ADDRESS MAP

Device	I^{2}C Address
OLED display	0x3C
ADS1115 (PL, PR)	0x48
Left MPU6050 (ML)	0x69
Right MPU6050 (MR)	0x68

IV. DATA ACQUISITION METHODOLOGY

The firmware acquires all sensor readings locally at a fixed interval before any data leaves the device, decoupling fast local sensing from slower cloud reporting.

A. Local Sensor Loop

Sensors are polled every 100 ms, corresponding to a 10 Hz acquisition rate. On each cycle, the ADS1115 is read in single ended mode to obtain raw piezo values for PL and PR, which are converted to voltage, while each MPU6050 reports acceleration along the X, Y, and Z axes. To obtain an orientation independent measure of movement, the acceleration magnitude is computed for each inertial sensor as

$$M = \sqrt{X^2 + Y^2 + Z^2}, \tag{1}$$

giving an ML magnitude and an MR magnitude on every cycle. Equation (1) is applied independently to the left and right MPU6050 readings.

B. Local Feedback

The current PL and PR readings and the ML and MR magnitudes, along with the WiFi connection status, are written to the onboard OLED display on every acquisition cycle, and the same values are also written to the serial console at 115200 baud. Fig. 3 shows the OLED in operation, confirming that the system provides immediate local feedback without requiring a connected computer or cloud dashboard. This local feedback path is intended to be useful during development and bench testing, before an onboard kick detection algorithm is implemented.

C. Cloud Upload

Independently of the 10 Hz local loop, the ESP32-S3 checks its WiFi connection and, once every 20 seconds, uploads a summarized set of readings to a ThingSpeak channel over HTTPS. If the WiFi link is found to be disconnected at upload

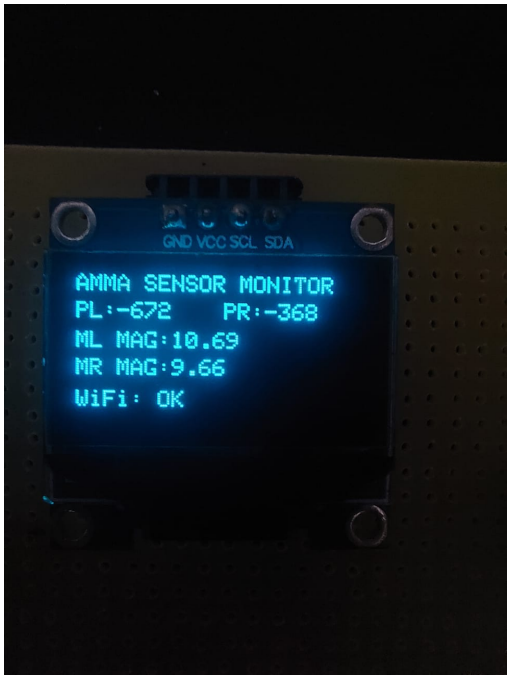


Fig. 3. OLED status display showing live PL, PR, ML magnitude, MR magnitude, and WiFi connection status.

time, the firmware attempts reconnection before the upload is skipped or retried. Table III lists the mapping between ThingSpeak channel fields and the AMMA sensor parameters.

TABLE III
THINGSPEAK FIELD MAPPING

Field	Parameter
1	PL raw ADC value
2	PR raw ADC value
3	ML acceleration magnitude
4	MR acceleration magnitude
5	ML X acceleration
6	ML Y acceleration
7	MR X acceleration
8	MR Y acceleration

V. RESULTS

The complete sensing to cloud pipeline was tested experimentally rather than only bench simulated. The ESP32-S3 successfully connected to the local WiFi network and obtained an IP address, after which uploads to the ThingSpeak channel returned an HTTP response code of 200 together with steadily incrementing entry identifiers, confirming that each upload was accepted by the cloud service rather than silently failing. Fig. 4 and Fig. 5 show the resulting ThingSpeak field charts for the piezo raw values and inertial readings respectively, captured across several logging sessions.

Because the local sensor loop runs at 10 Hz while the cloud upload occurs roughly every 20 seconds, the ThingSpeak charts should be read as a trend and dataset collection view of device behavior rather than a real time waveform of individual



Fig. 4. ThingSpeak charts for Fields 1 to 4: PL raw, PR raw, ML acceleration magnitude, and MR acceleration magnitude.



Fig. 5. ThingSpeak charts for Fields 5 to 8: ML X, ML Y, MR X, and MR Y acceleration components.

movements; fine grained event level detection is intended to happen locally on the ESP32-S3 in a later development stage. A mobile dashboard mock-up illustrating how these readings, together with derived measures such as fetal kick counts, heart rate, and SpO2, are intended to be presented to the end user is shown in Fig. 6. This dashboard view is a design mock-up used to guide the companion application and is not yet connected to live AMMA sensor data.

VI. CURRENT STATUS AND LIMITATIONS

At the present stage, the hardware integration, the 100 ms sensor acquisition loop, the OLED and serial feedback paths, and the WiFi to ThingSpeak upload pipeline have all been implemented and verified to function together. The system should be described as a functional multi-sensor maternal movement monitoring and cloud data acquisition prototype; it does not yet perform automated fetal kick detection or AI based classification, and these capabilities should not be claimed at this stage of the project. Fig. 7 outlines the intended remaining pipeline, proceeding from raw sensor data through signal preprocessing, noise filtering, baseline estimation, and

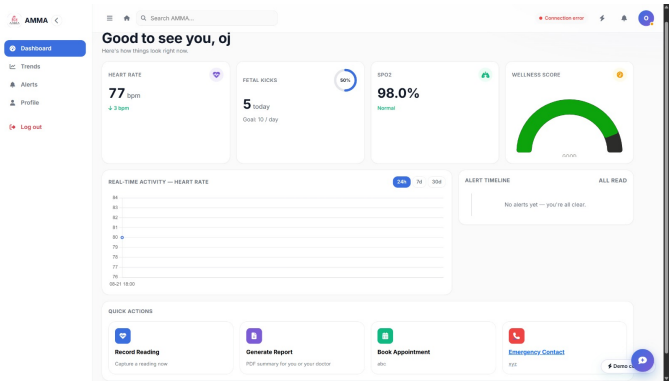


Fig. 6. Mock-up of the planned AMMA companion dashboard, showing heart rate, fetal kick count, SpO2, and a composite wellness score.

movement or event detection, to feature extraction, kick event classification, and finally kick count and maternal insight generation surfaced through a mobile application together with AI assisted analysis.

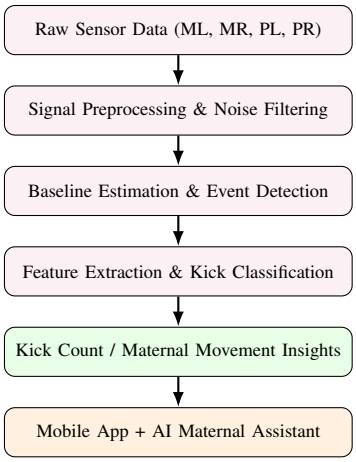


Fig. 7. Planned future processing pipeline, extending the current data acquisition prototype toward automated kick detection and AI assisted maternal insights. This pipeline is not yet implemented.

Sensor accuracy in the presence of motion artefacts, the need for larger scale data collection prior to model training, and dependence on consistent user compliance for continuous wear are recognized as open constraints to be addressed in subsequent phases.

VII. Conclusion and Future Work

This paper reported the current AMMA prototype: a dual MPU6050, dual piezoelectric sensor wearable platform built on an ESP32-S3, with a verified 10 Hz local acquisition loop, OLED and serial feedback, and a working WiFi based upload to a ThingSpeak cloud channel confirmed by successful HTTP responses and incrementing entry identifiers. The system establishes the hardware and data acquisition foundation on which the next phase of the project will be built. Future work will focus on local signal preprocessing and noise filtering, development of a movement and kick event detection algorithm, feature extraction and classification for distinguishing normal from abnormal fetal movement patterns, and integration of these results into a mobile application and AI based maternal assistant, moving the project from data acquisition toward the intelligent monitoring capability that is the ultimate goal of the AMMA Maternal Wellness Suite.

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